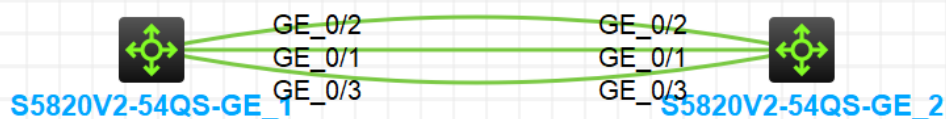


链路聚合实验

实验拓扑

实验需求

- 1.在 **SW1** 和 **SW2** 之间配置动态链路聚合，允许所有 **VLAN** 通过
- 2.要求 **SW1** 和 **SW2** 之间的最大传输带宽为 **2G**



实验需求

1. 在 SW1 和 SW2 之间配置动态链路聚合，允许所有 VLAN 通过
2. 要求 SW1 和 SW2 之间的最大传输带宽为 2G

实验配置

```
1 [SW1]int Bridge-Aggregation 1
2 [SW1-Bridge-Aggregation1]qu
3 [SW1]int range GigabitEthernet 1/0/1 to GigabitEthernet 1/0/3
4 [SW1-if-range]port link-aggregation group 1
5 [SW1]int Bridge-Aggregation 1
6 [SW1-Bridge-Aggregation1]port link-type trunk
7 Configuring GigabitEthernet1/0/1 done.
8 Configuring GigabitEthernet1/0/2 done.
9 Configuring GigabitEthernet1/0/3 done.
10 [SW1-Bridge-Aggregation1]port trunk permit vlan all
11 Configuring GigabitEthernet1/0/1 done.
12 Configuring GigabitEthernet1/0/2 done.
13 Configuring GigabitEthernet1/0/3 done.
14 [SW1-Bridge-Aggregation1]link-aggregation mode dynamic
15 [SW1-Bridge-Aggregation1]link-aggregation selected-port maximum 2
16 ---
17 [SW2]int Bridge-Aggregation 1
18 [SW2-Bridge-Aggregation1]qu
19 [SW2]int range GigabitEthernet 1/0/1 to GigabitEthernet 1/0/3
20 [SW2-if-range]port link-aggregation group 1
21 [SW2]interface Bridge-Aggregation 1
22 [SW2-Bridge-Aggregation1]port link-type trunk
23 Configuring GigabitEthernet1/0/1 done.
24 Configuring GigabitEthernet1/0/2 done.
```

```

25 Configuring GigabitEthernet1/0/3 done.
26 [SW2-Bridge-Aggregation1]port trunk permit vlan a
27 Configuring GigabitEthernet1/0/1 done.
28 Configuring GigabitEthernet1/0/2 done.
29 Configuring GigabitEthernet1/0/3 done.
30 [SW2-Bridge-Aggregation1]link-aggregation mode dynamic
31 [SW2-Bridge-Aggregation1]link-aggregation selected-port maximum 2

```

结果验证

1.未设置最大可用端口时,查看链路聚合信息

```

1 [SW1-Bridge-Aggregation1]display link-aggregation summary
2 Aggregation Interface Type:
3 BAGG -- Bridge-Aggregation, BLAGG -- Blade-Aggregation, RAGG -- Route-Aggregation, SCH-B --
  Schannel-Bundle
4 Aggregation Mode: S -- Static, D -- Dynamic
5 Loadsharing Type: Shar -- Loadsharing, NonS -- Non-Loadsharing
6 Actor System ID: 0x8000, 48e2-67b9-0100
7
8 AGG      AGG  Partner ID          Selected  Unselected  Individual  Share
9 Interface Mode                                     Ports      Ports        Ports        Type
10 -----
11 BAGG1    D    0x8000, 48e2-689d-0200  3         0           0           Shar
12 [SW1-Bridge-Aggregation1]display link-aggregation verbose
13 Loadsharing Type: Shar -- Loadsharing, NonS -- Non-Loadsharing
14 Port: A -- Auto
15 Port Status: S -- Selected, U -- Unselected, I -- Individual
16 Flags: A -- LACP_Activity, B -- LACP_Timeout, C -- Aggregation,
17        D -- Synchronization, E -- Collecting, F -- Distributing,
18        G -- Defaulted, H -- Expired
19
20 Aggregate Interface: Bridge-Aggregation1
21 Aggregation Mode: Dynamic
22 Loadsharing Type: Shar
23 System ID: 0x8000, 48e2-67b9-0100
24 Local:
25   Port          Status  Priority Oper-Key  Flag
26   -----
27   GE1/0/1       S       32768   1        {ACDEF}
28   GE1/0/2       S       32768   1        {ACDEF}
29   GE1/0/3       S       32768   1        {ACDEF}
30 Remote:
31   Actor          Partner Priority Oper-Key  SystemID          Flag
32   -----
33   GE1/0/1       2       32768   1        0x8000, 48e2-689d-0200 {ACDEF}
34   GE1/0/2       3       32768   1        0x8000, 48e2-689d-0200 {ACDEF}
35   GE1/0/3       4       32768   1        0x8000, 48e2-689d-0200 {ACDEF}

```

2.在设置了最大可用端口并shutdown SW1的1/0/1端口时查看链路聚合信息

```

1 [SW1-GigabitEthernet1/0/1]shutdown

```

```

2 [SW1-GigabitEthernet1/0/1]display link-aggregation summary
3 Aggregation Interface Type:
4 BAGG -- Bridge-Aggregation, BLAGG -- Blade-Aggregation, RAGG -- Route-Aggregation, SCH-B --
  Schannel-Bundle
5 Aggregation Mode: S -- Static, D -- Dynamic
6 Loadsharing Type: Shar -- Loadsharing, NonS -- Non-Loadsharing
7 Actor System ID: 0x8000, 48e2-67b9-0100
8
9 AGG      AGG  Partner ID          Selected  Unselected  Individual  Share
10 Interface Mode                               Ports      Ports        Ports        Type
11 -----
12 BAGG1    D    0x8000, 48e2-689d-0200  2         1           0           Shar
13 [SW1-GigabitEthernet1/0/1]display link-aggregation verbose
14 Loadsharing Type: Shar -- Loadsharing, NonS -- Non-Loadsharing
15 Port: A -- Auto
16 Port Status: S -- Selected, U -- Unselected, I -- Individual
17 Flags: A -- LACP_Activity, B -- LACP_Timeout, C -- Aggregation,
18        D -- Synchronization, E -- Collecting, F -- Distributing,
19        G -- Defaulted, H -- Expired
20
21 Aggregate Interface: Bridge-Aggregation1
22 Aggregation Mode: Dynamic
23 Loadsharing Type: Shar
24 System ID: 0x8000, 48e2-67b9-0100
25 Local:
26   Port                Status  Priority Oper-Key  Flag
27   -----
28   GE1/0/1             U      32768   1        {AC}
29   GE1/0/2             S      32768   1        {ACDEF}
30   GE1/0/3             S      32768   1        {ACDEF}
31 Remote:
32   Actor                Partner Priority Oper-Key  SystemID          Flag
33   -----
34   GE1/0/1             2      32768   1        0x8000, 48e2-689d-0200 {ACEF}
35   GE1/0/2             3      32768   1        0x8000, 48e2-689d-0200 {ACDEF}
36   GE1/0/3             4      32768   1        0x8000, 48e2-689d-0200 {ACDEF}
37 ---
38 <SW2>display link-aggregation summary
39 Aggregation Interface Type:
40 BAGG -- Bridge-Aggregation, BLAGG -- Blade-Aggregation, RAGG -- Route-Aggregation, SCH-B --
  Schannel-Bundle
41 Aggregation Mode: S -- Static, D -- Dynamic
42 Loadsharing Type: Shar -- Loadsharing, NonS -- Non-Loadsharing
43 Actor System ID: 0x8000, 48e2-689d-0200
44
45 AGG      AGG  Partner ID          Selected  Unselected  Individual  Share
46 Interface Mode                               Ports      Ports        Ports        Type
47 -----
48 BAGG1    D    0x8000, 48e2-67b9-0100  2         1           0           Shar
49 <SW2>display link-aggregation verbose
50 Loadsharing Type: Shar -- Loadsharing, NonS -- Non-Loadsharing
51 Port: A -- Auto
52 Port Status: S -- Selected, U -- Unselected, I -- Individual

```

```

53 Flags:  A -- LACP_Activity, B -- LACP_Timeout, C -- Aggregation,
54         D -- Synchronization, E -- Collecting, F -- Distributing,
55         G -- Defaulted, H -- Expired
56
57 Aggregate Interface: Bridge-Aggregation1
58 Aggregation Mode: Dynamic
59 Loadsharing Type: Shar
60 System ID: 0x8000, 48e2-689d-0200
61 Local:
62   Port                Status  Priority Oper-Key  Flag
63   -----
64   GE1/0/1             U      32768    1        {AC}
65   GE1/0/2             S      32768    1        {ACDEF}
66   GE1/0/3             S      32768    1        {ACDEF}
67 Remote:
68   Actor                Partner Priority Oper-Key  SystemID          Flag
69   -----
70   GE1/0/1             2      32768    1        0x8000, 48e2-67b9-0100 {ACEF}
71   GE1/0/2             3      32768    1        0x8000, 48e2-67b9-0100 {ACDEF}
72   GE1/0/3             4      32768    1        0x8000, 48e2-67b9-0100 {ACDEF}

```